

ComicCraft – AI Comic Story Creator

Phase 2: Requirement Analysis Phase

1. Introduction

Requirement Analysis identifies what the ComicCraft application must provide to transform user prompts into personalized comic stories and illustrations. The requirements are derived from the provided project description, scenarios, architecture, and workflow.

2. Main Project Requirement

The application must accept user-provided comic information and generate a complete comic experience containing a structured storyline, narration, character dialogue, illustrations, preview, and downloadable PDF.

3. User-Centered Requirement

The interface should allow users to enter a story prompt, main character name, setting, story tone, and preferred art style. The generated result should be displayed in an accessible web interface.

Phase 2 – Functional Requirements

1. User Input

The system must collect the Story Prompt, Character Name, Setting, Story Tone, and Art Style. The frontend form sends these details to the FastAPI backend for processing.

2. Comic Outline Generation

The system requires a structured comic outline. The project specifies a five-panel outline, where each panel contains a panel number, title, scene description, and image-generation prompt.

3. Story and Dialogue Generation

The system must expand the panel outline into a full comic-style story with narration and character dialogues for each panel.

4. Illustration Generation

For every panel, the system must generate a comic-style illustration using the provided image prompt and save the resulting image for use in the comic layout.

Phase 2 – Processing & Integration Requirements

1. Backend Processing

FastAPI is required to receive form data or JSON requests, process the input, coordinate AI generation, organize the comic, and provide the final result to the frontend.

2. AI Integration

Gemini Flash is specified for structured panel outlines, Gemini Pro for detailed narration and dialogue, and Stable Diffusion for comic-style illustrations.

3. Comic Layout

The generated images and story text must be matched panel-by-panel and organized into a structured layout before preview and export.

4. API Requirements

The documented application includes routes for browser-based generation, JSON-based comic generation, image testing, and export-success handling.

Phase 2 – Interface & Output Requirements

1. Homepage Requirement

The homepage must provide an input form for the story prompt, character name, setting, tone, and art style. It acts as the starting point for comic creation.

2. Comic Preview Requirement

The preview page must display the generated comic panel-by-panel. Each panel can contain its title, illustration, scene description, captions, and narration.

3. PDF Export Requirement

The system must compile the comic images and narrative content into a multi-page PDF. The project specifies FPDF for PDF generation and a downloadable export flow.

4. Success Page

After the comic is exported, the application must provide a success confirmation and allow the user to return and create another comic.

Phase 2 – Technical & Deployment Requirements

1. Technology Requirements

The project identifies Python, FastAPI, Uvicorn, HTML, CSS, Jinja2, Google Gemini APIs, Hugging Face Diffusers, Stable Diffusion, PyTorch, FPDF, and Git as relevant technologies or prerequisites.

2. Environment Requirements

The application requires a Python environment with the required packages installed. A virtual environment is recommended in the project workflow to isolate project dependencies.

3. Security Configuration

Sensitive credentials such as the Gemini API key and Hugging Face API token are specified to be stored through environment variables or a .env configuration.

4. Testing & Verification

The application must be tested locally using Uvicorn. The documented workflow includes access to the application at the local server and FastAPI's /docs interface for API testing.

5. Requirement Analysis Conclusion

The requirements define an end-to-end AI comic generation system: collect creative input, generate structured story content, create illustrations, build the comic layout, preview it, and export the final comic as a PDF.